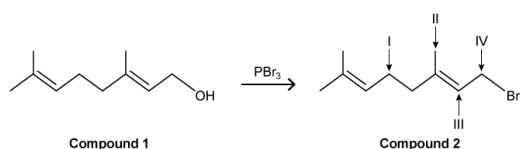


Compound 1 and PBr_3 react to form Compound 2 as shown below.



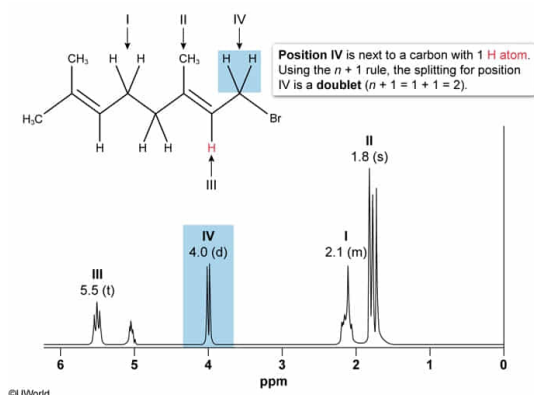
Which position in Compound 2 corresponds to the hydrogen(s) that appear as a doublet at approximately 4.0 ppm in the ^1H NMR spectrum?

- ☐ A. I [13%]
☐ B. II [10%]
☐ C. III [23%]
☒ D. IV [52%]

Correct

52% answered correctly

Explanation:



Proton **nuclear magnetic resonance spectroscopy** (^1H NMR) detects hydrogen atoms (^1H isotope) in a molecule as an external magnetic field and radio frequency pulse are applied to the sample. The hydrogen atoms in different magnetic environments (resulting from molecular structure) are nonequivalent, and neighboring nonequivalent hydrogens within three bonds of each other cause **spin-spin splitting** of the peaks in the NMR spectrum. The **splitting pattern** can be determined by **$n + 1$ rule**, where n is the number of nonequivalent hydrogen atoms on adjacent carbons.

Position IV in Compound 2 contains two hydrogen atoms that are adjacent to a single H atom bonded to an sp^2 carbon. According to the $n + 1$ rule, $n = 1$ because there is only one coupled hydrogen to the hydrogen atoms at position IV. Therefore, because $n + 1 = 2$, the peak will be split in two, making the splitting pattern a doublet.

(Choice A) Position I contains two hydrogen atoms that are split by three adjacent hydrogen atoms. Although a quartet might be expected, complex, secondary splitting causes overlapping peaks to appear as a **multiplet** at 2.1 ppm.

(Choice B) Position II consists of a methyl group (three hydrogen atoms) that is adjacent to a

4° carbon (no bonds to hydrogen). Because there are no neighboring hydrogen atoms coupled to the methyl hydrogens, the peak is not split and appears as a **singlet** at 1.8 ppm.

(Choice C) Position III contains an alkene hydrogen atom that is adjacent to an alkyl ($-\text{CH}_2-$) group. The two alkyl hydrogen atoms are coupled to the alkene hydrogen, causing the peak to be split into a **triplet**.

Educational objective:

^1H NMR detects hydrogen atoms in a molecule as an external magnetic field and radio frequency pulse are applied to the sample. Hydrogen atoms in different magnetic environments within three bonds of each other cause spin-spin splitting of the peaks in the spectrum. The splitting pattern can be determined by the $n + 1$ rule (n = number of nonequivalent hydrogen atoms on adjacent carbons).

Time spent:0 sec

Subject:
Organic Chemistry

QID:401694

System:
Separation Techniques, Spectroscopy, and Analytical Methods